

Eigenvalues and inverse problems

Symmetric minimizer for a positive definite Jordan–Kronecker product

Difficulty: challenging **Importance:** interesting to the community

Status: [Lean verified](#)

Topic: Structured eigenvalue problems and semidefinite optimization.

Rating rationale: Challenging reflects a structured minimum-eigenvalue comparison for arbitrary positive definite pairs; community impact connects Kronecker eigenproblems and semidefinite optimization.

Lean verification — 2026-09-15

The complete original inequality is **Lean verified** for every dimension $n \geq 2$ and every real symmetric positive definite pair A, B . Both Rayleigh minima are attained over their full nonzero symmetric and skew sectors. The stronger source theorem supplies a real nonzero positive-semidefinite eigenmatrix at the global minimum of $A \otimes B + B \otimes A$ for every $n \geq 1$. All six independently reviewed targets passed [fresh Linux Comparator and default-kernel verification](#).

Formalization: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, with substantial OpenAI Codex assistance. **Original mathematical proof:** Matthew J. Colbrook, as credited below. The original conjecture remains credited to N. Kalantarova and L. Tunçel. [Lean project and reproduction](#) · [formalization metadata](#) · [two source and two operational reviews](#).

Resolution — 2026-09-11

Affirmative resolution by Matthew J. Colbrook (Department of Applied Mathematics and Theoretical Physics, University of Cambridge). See the [complete manuscript](#), **Theorem SP-05, sections 1–3** (PDF · [LaTeX](#)), prepared 11 September 2026.

For arbitrary real symmetric positive definite A, B , a nonzero real positive-semidefinite eigenmatrix attains the smallest eigenvalue of $X \mapsto AXB + BXA$. Section 3 derives the exact symmetric/skew-symmetric Rayleigh-quotient inequality in the original target, without commutativity, rank restrictions or a simple-eigenvalue assumption.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently verified the full proof and its match to the exact target on 11 September 2026: [detailed PASS review](#). The review records a hash of the unchanged proof text. This is independent agent verification, not external human peer review or formal certification. [The submission history and diagnostic record](#) preserve the initial solution claim. The ratings above are historical, and the earlier literature checks below are retained.

Problem statement

Let $n \geq 2$ and let $A, B \in \mathbb{R}^{n \times n}$ be symmetric positive definite. Let $T \in \mathbb{R}^{n^2 \times n^2}$ be the commutation matrix defined by $T \operatorname{vec}(X) = \operatorname{vec}(X^T)$, where vec stacks columns. Is it always true that

$$\min_{\substack{u \in \mathbb{R}^{n^2} \setminus \{0\} \\ Tu = u}} \frac{u^T (A \otimes B) u}{u^T u} \leq \min_{\substack{w \in \mathbb{R}^{n^2} \setminus \{0\} \\ Tw = -w}} \frac{w^T (A \otimes B) w}{w^T w}?$$

Equivalently, must the smallest eigenvalue of $A \otimes B + B \otimes A$ have an eigenvector $\operatorname{vec}(X)$ with $X = X^T \neq 0$?

Why it matters

An affirmative answer would identify which invariant subspace contains the smallest eigenvalue of an important structured matrix operator, relevant to positivity checks in semidefinite optimization.

References

- N. Kalantarova and L. Tunçel, [On the spectral structure of Jordan–Kronecker products of symmetric and skew-symmetric matrices](#), *Linear Algebra and its Applications* 608 (2021), 343–362. Conjecture 1, equation (7), p. 12 of arXiv v3; the preceding discussion explains the optimization connection.
- D. Kressner and B. Vandereycken, [A counterexample to the symmetric-maximizer conjecture for Lyapunov operators](#), 2026, abstract. This resolves a different symmetric-maximizer conjecture and is included to distinguish the recent result from this problem.

Earlier status check — 2026-09-08

On 2026-09-08, checked arXiv v3 (2020-08-08, still the latest listed version) and searched combinations of “Jordan-Kronecker”, “positive definite”, “conjecture”, “smallest”, “counterexample”, and the authors’ names, including 2025–2026 results. No resolution of Conjecture 1 was located. The source’s counterexamples to general symmetric-matrix interlacing do not dispose of its separately stated positive definite conjecture. The August 2026 Lyapunov result concerns an operator norm maximizer, rather than this positive definite Jordan–Kronecker minimum. No recent explicit reaffirmation of this exact conjecture was found; the status evidence is therefore bounded.

Audit update — 2026-09-10

Rechecked the [author manuscript](#), Conjecture 1 and equation (7). Its positive definite, minimum-eigenvalue formulation matches this entry. Searches for the Jordan–Kronecker symmetric-minimizer conjecture and later counterexamples found no resolution. Results for indefinite inputs or a maximum-eigenvalue objective do not settle the displayed claim.